Methods in Human Geography

Quantitative Methods: Statistical Analysis II



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This week

Part I

- Crosstabulation.
- Correlation.

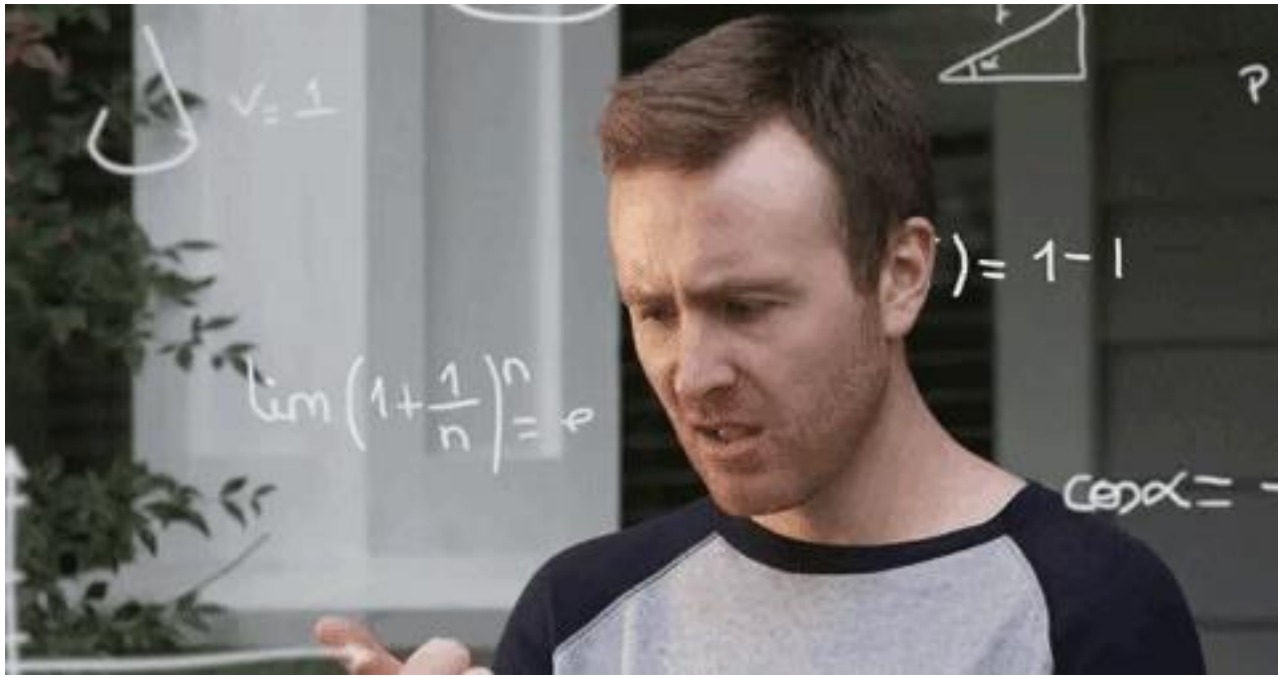
Part II

- Regression.
- Assumptions.

Choosing a method

Dependent variable (y)	Independent variable (x)	Method	Non-parametric alternative
Continuous	Categorical, two groups	t-test	Mann-Whitney U test
Continuous	Categorical, three or more groups	ANOVA	Kruskal-Wallis test
Categorical	Categorical	Crosstabulation and chi-square test	–
Continuous	Continuous	Correlation and regression	Spearman's correlation

Last week: the first two rows. This week: the last two rows.


V= 1
lim (1+1/n)^n = e
= 1-1

Crosstabulation

Crosstabulation

- We often encounter nominal or ordinal variables like gender, ethnic group, educational qualifications, income bands, or age groups.
- Information across two variables can be presented with a crosstab (contingency table) that organises data into a matrix, showing the frequency distribution across categories of two categorical variables.

Crosstabulation

	More than 40% aged 50 or over		
Winner	**No**	**Yes**	***Total***
Conservative	25	91	116
Labour	243	131	374
Liberal Democrats	24	42	66
Other	8	11	19
Total	300	275	575

Rows: the winning party. Columns: whether more than 40% of residents are aged 50 or over.

Crosstabulation

	More than 40% aged 50 or over		
Winner	**No**	**Yes**	***Total***
Conservative	25 (21.6%)	91 (78.4%)	116 (20.2%)
Labour	243 (65.0%)	131 (35.0%)	374 (65.0%)
Liberal Democrats	24 (36.4%)	42 (63.6%)	66 (11.5%)
Other	8 (42.1%)	11 (57.9%)	19 (3.3%)
Total	300	275	575

Conservative seats: 78.4% have many over-50s. Labour seats: 35.0%.

Chi-square test

	More than 40% aged 50 or over		
Winner	**No**	**Yes**	***Total***
Conservative	25 60.5	91 55.5	116
Labour	243 195.1	131 178.9	374
Liberal Democrats	24 34.4	42 31.6	66
Other	8 9.9	11 9.1	19
Total	300	275	575

Red: the counts we would expect if party and age were unrelated (the null hypothesis).

Chi-square test

- Chi-square test (χ^2) assesses if the observed frequencies differ significantly from expected frequencies:

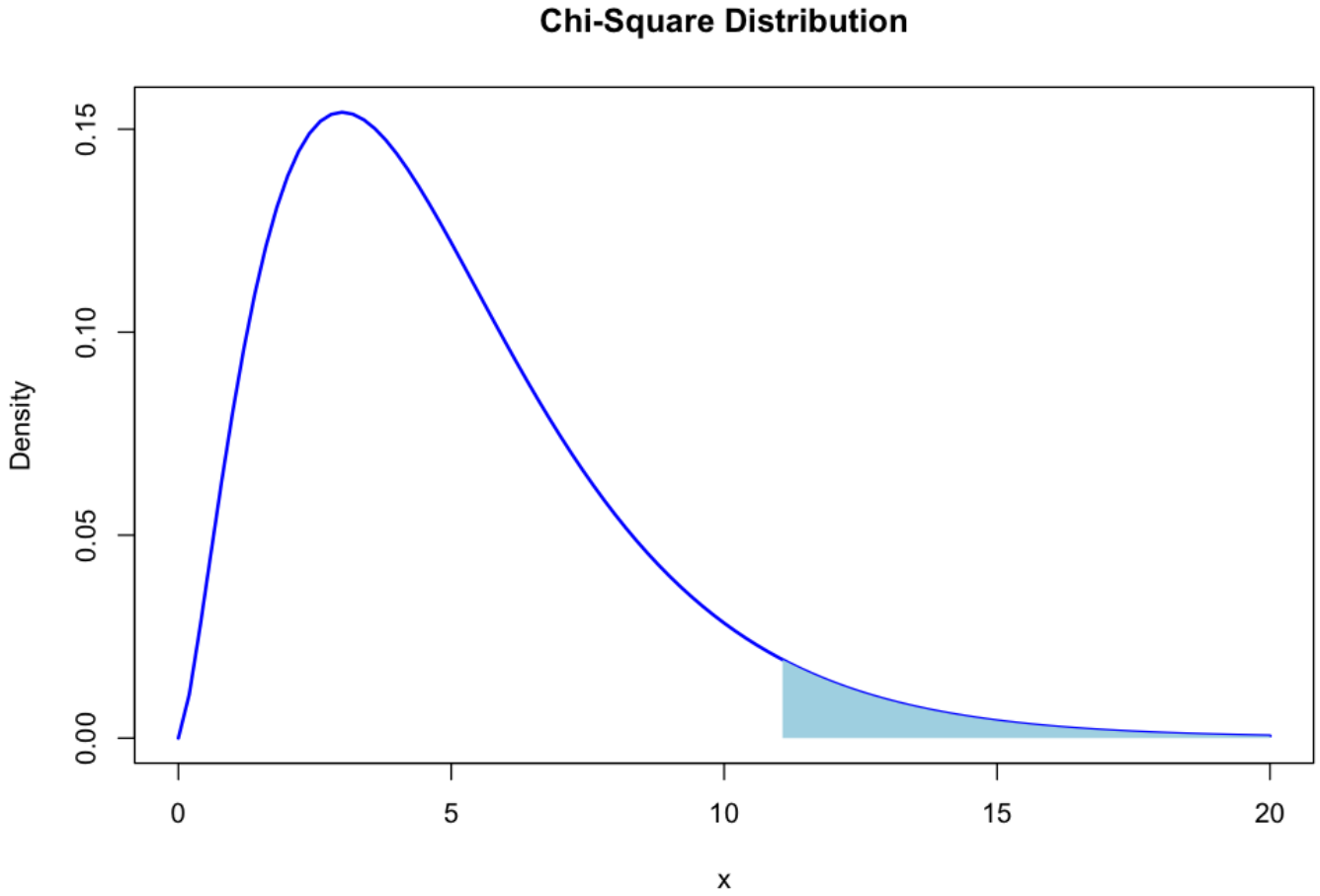
$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

- A significant chi-square result suggests an association, while a non-significant result means we found no evidence of an association.
- Example, Conservative and no: (25 - 60.5)² / 60.5 = 20.8. Summing all eight cells gives a chi-square statistic of 75.5.

Chi-square test

- Chi-square requires a sufficiently large sample size and expected frequency of at least five in each cell for valid results.
- It tells us whether an association is unlikely to be due to chance, but not how strong it is. For the size of the difference, compare the actual percentages: 78.4% against 35.0% in our example.

Chi-square test



Ecological fallacy

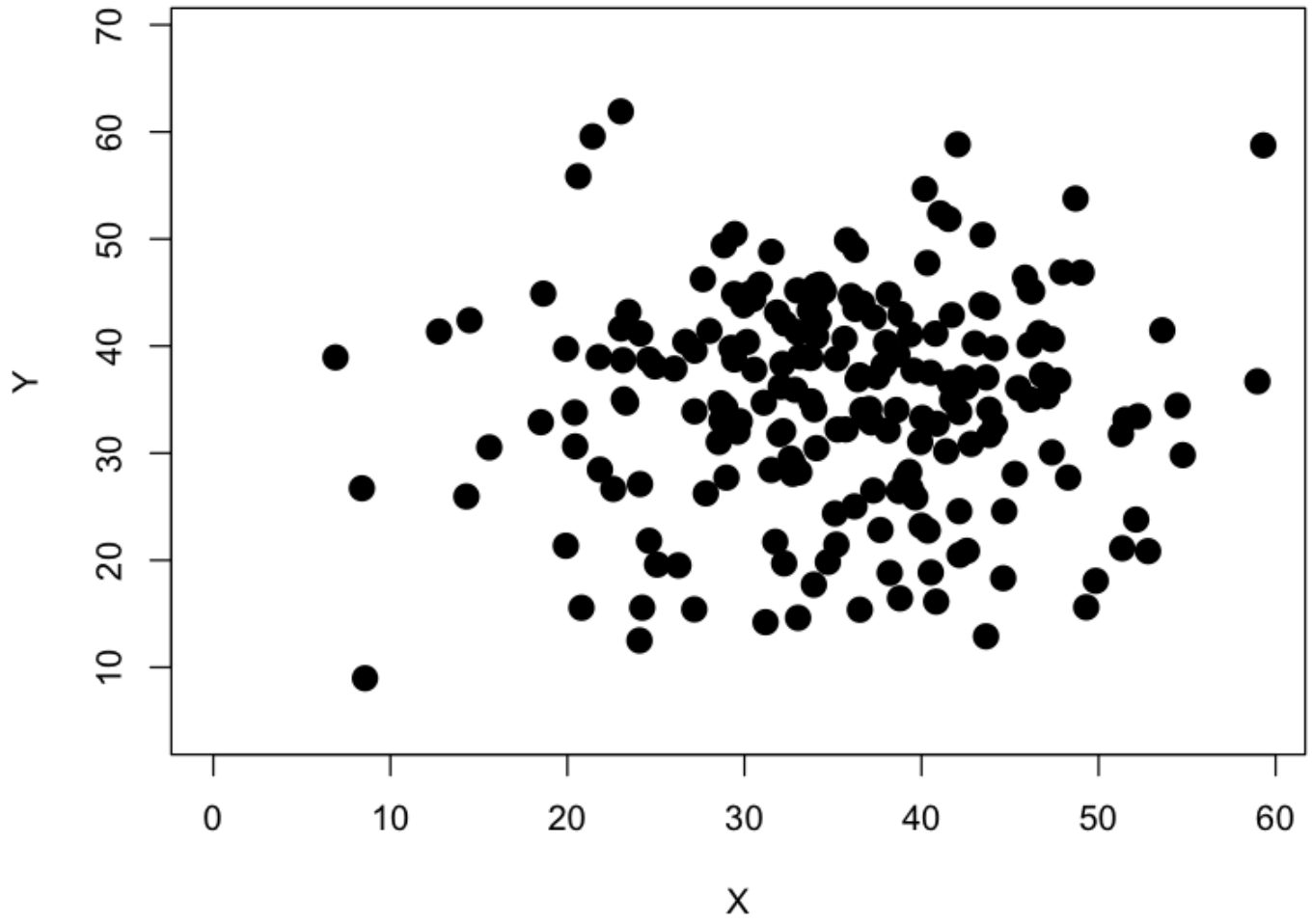
- Our data describe constituencies, not individual voters.
- Constituencies with an older population are more likely to be won by the Conservatives. This does not show that older people vote Conservative.
- Drawing conclusions about individuals from data about areas is what is called the ecological fallacy.
- The unit of analysis decides what we can say: data about areas support statements about areas.

Correlation

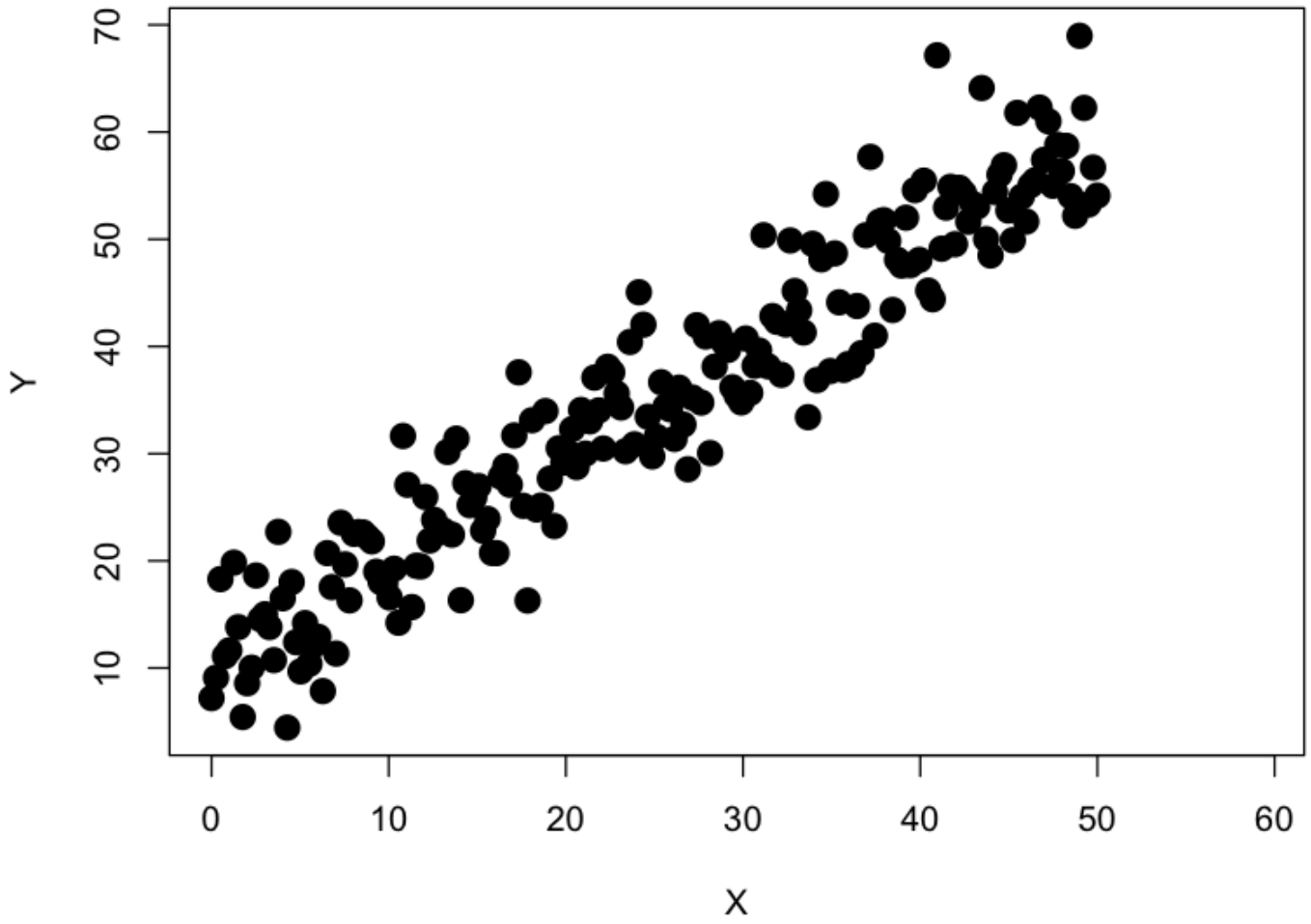
Scatterplot

- Crosstabulations are useful for examining bivariate relationships between nominal and ordinal data – but what about continuous data?
- Scatterplots offer a visual representation of the relationship between two continuous variables, though interpretation can be subjective.

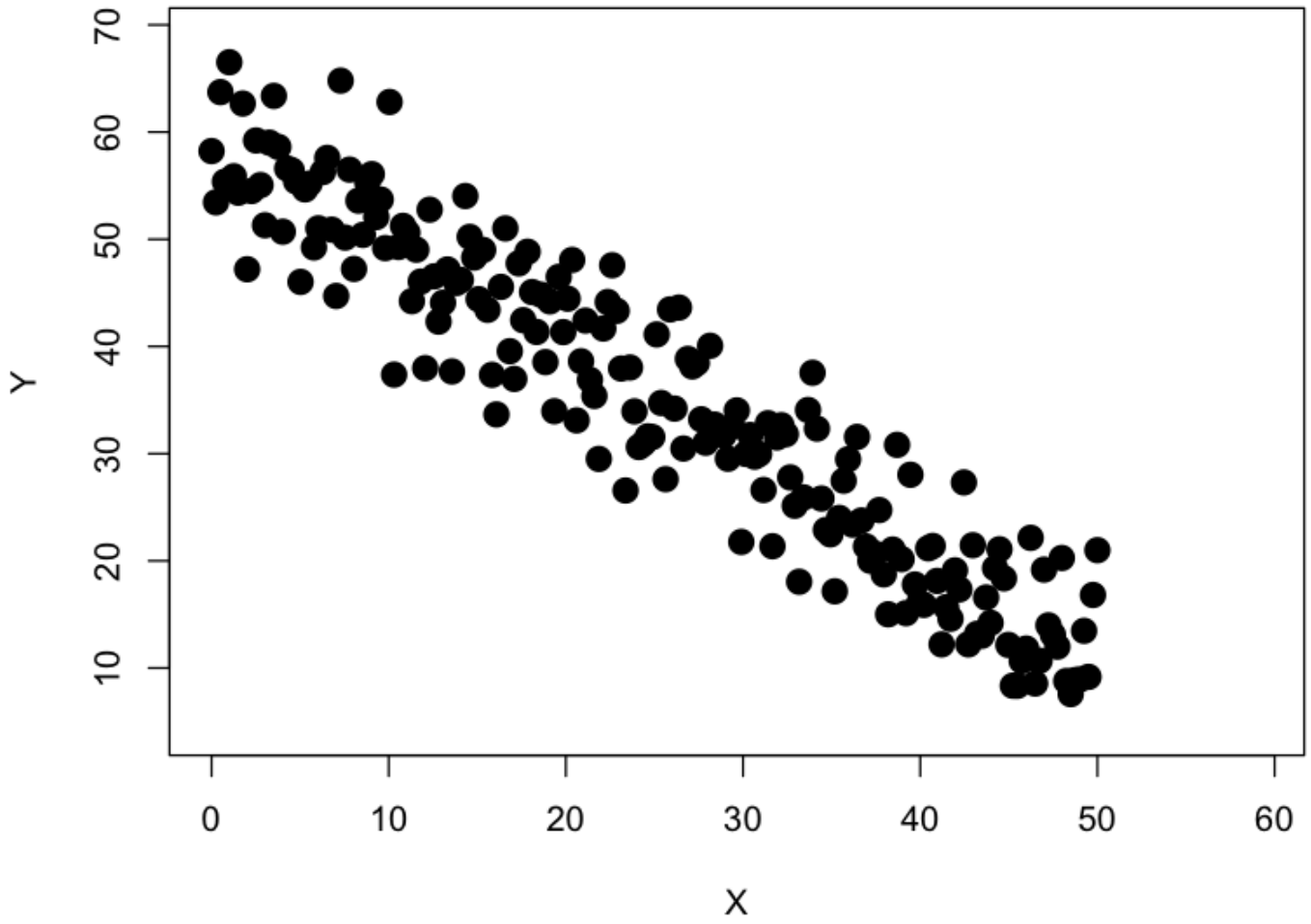
Correlation



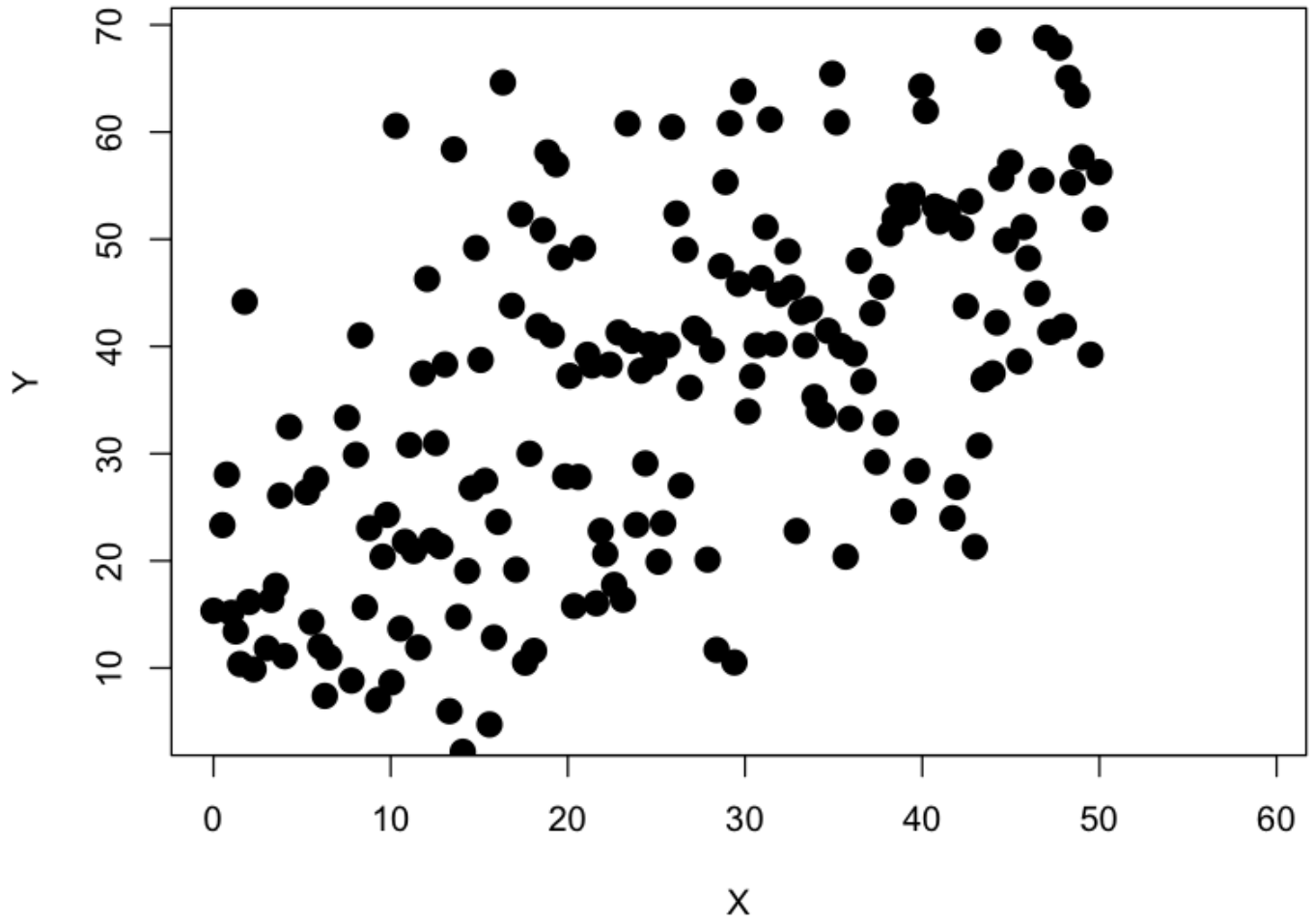
Correlation



Correlation



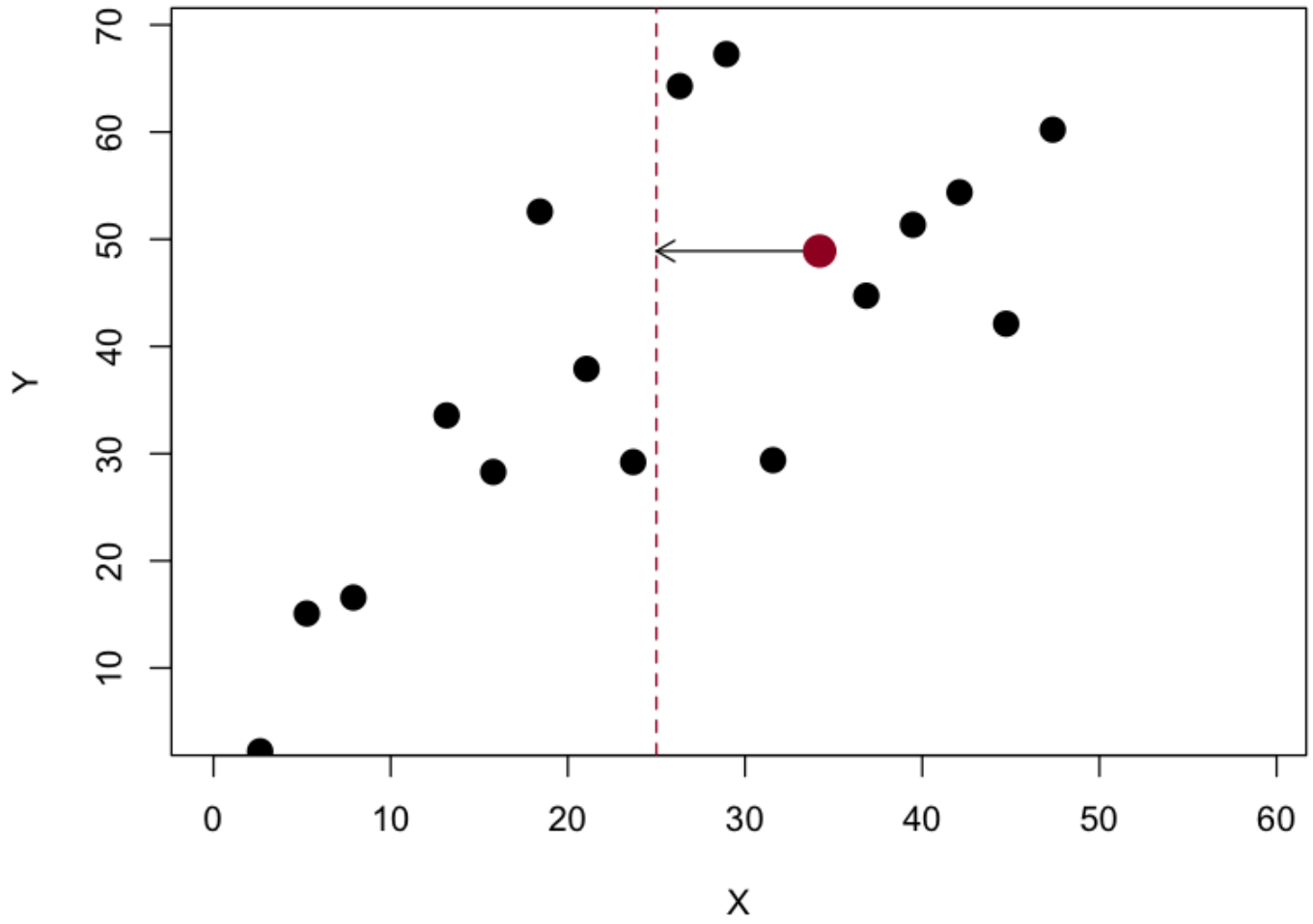
Correlation



Correlation

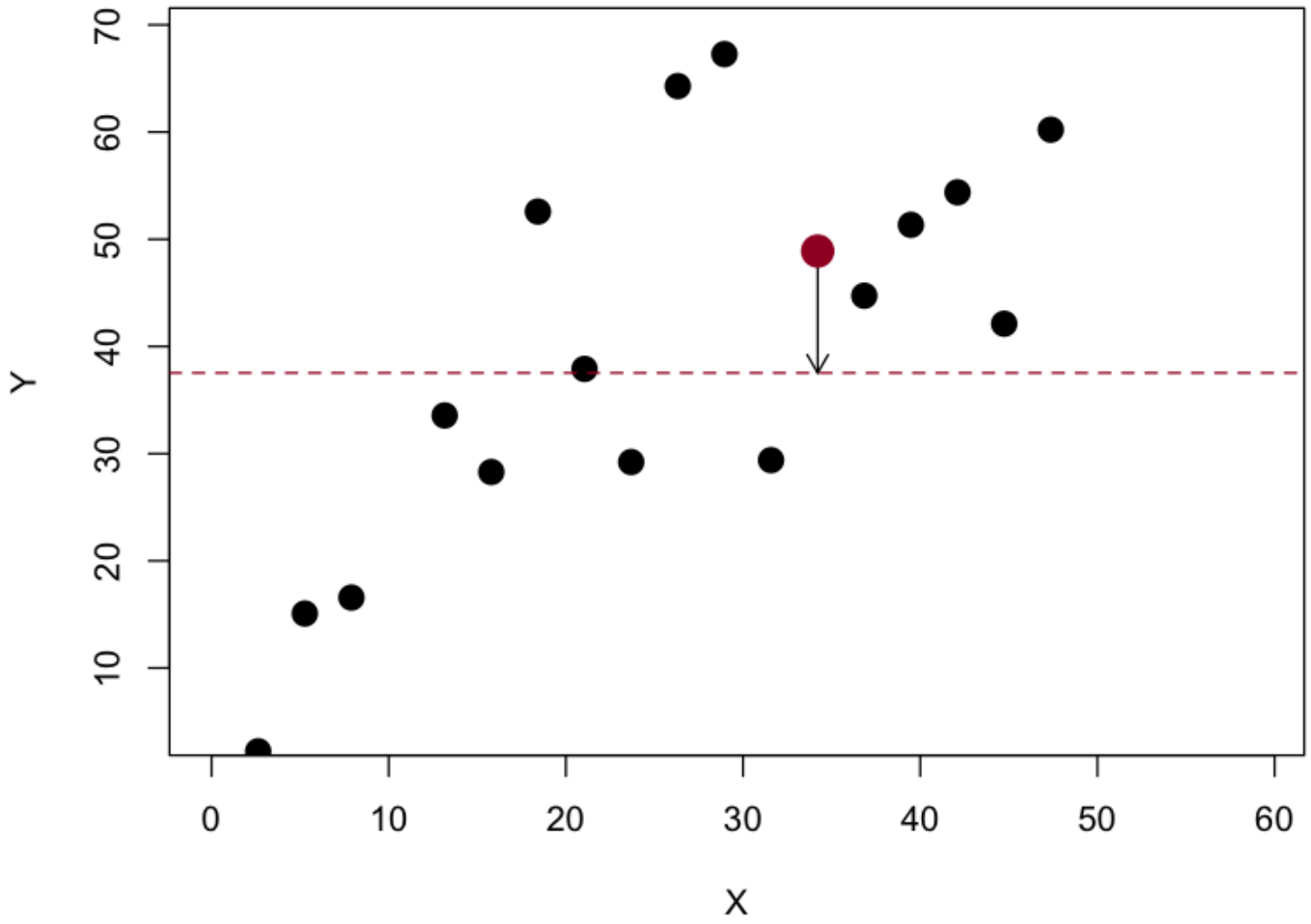
- Correlation is the degree to which two variables ‘vary together’ or are ‘related’.
- Variables are correlated when there is some change in one variable at the same time as there is a change in another variable.
- Correlation quantifies both the strength and direction of the relationship between two variables.
- There are several measures of correlation, with Pearson’s correlation coefficient being the most commonly used.

Variance



Variance: the average squared distance from the mean of x.

Variance



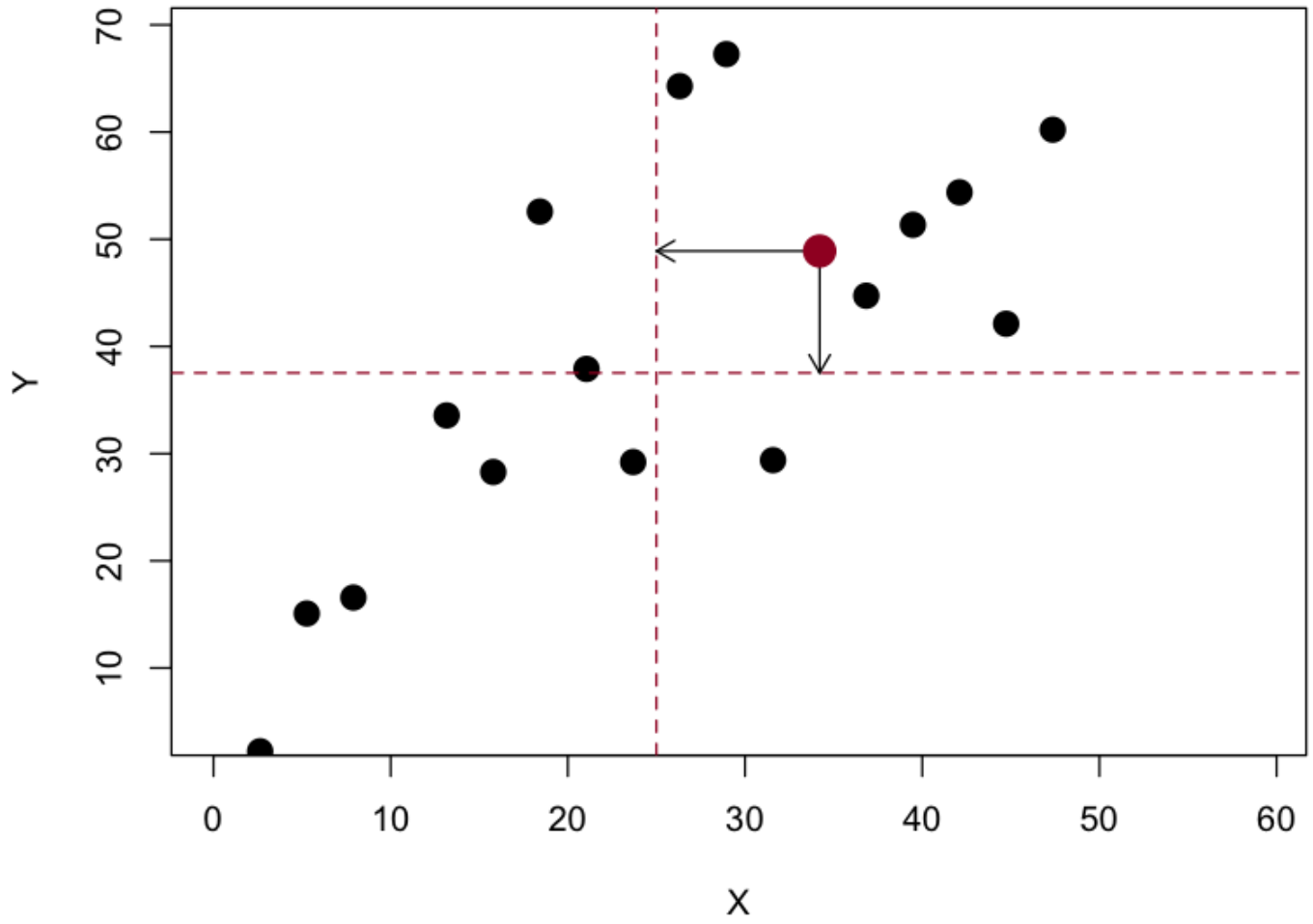
Variance: the average squared distance from the mean of y.

Variance

- Variance is a statistical measure that quantifies the dispersion of a variable's values around its mean, indicating how much the values differ from the average:

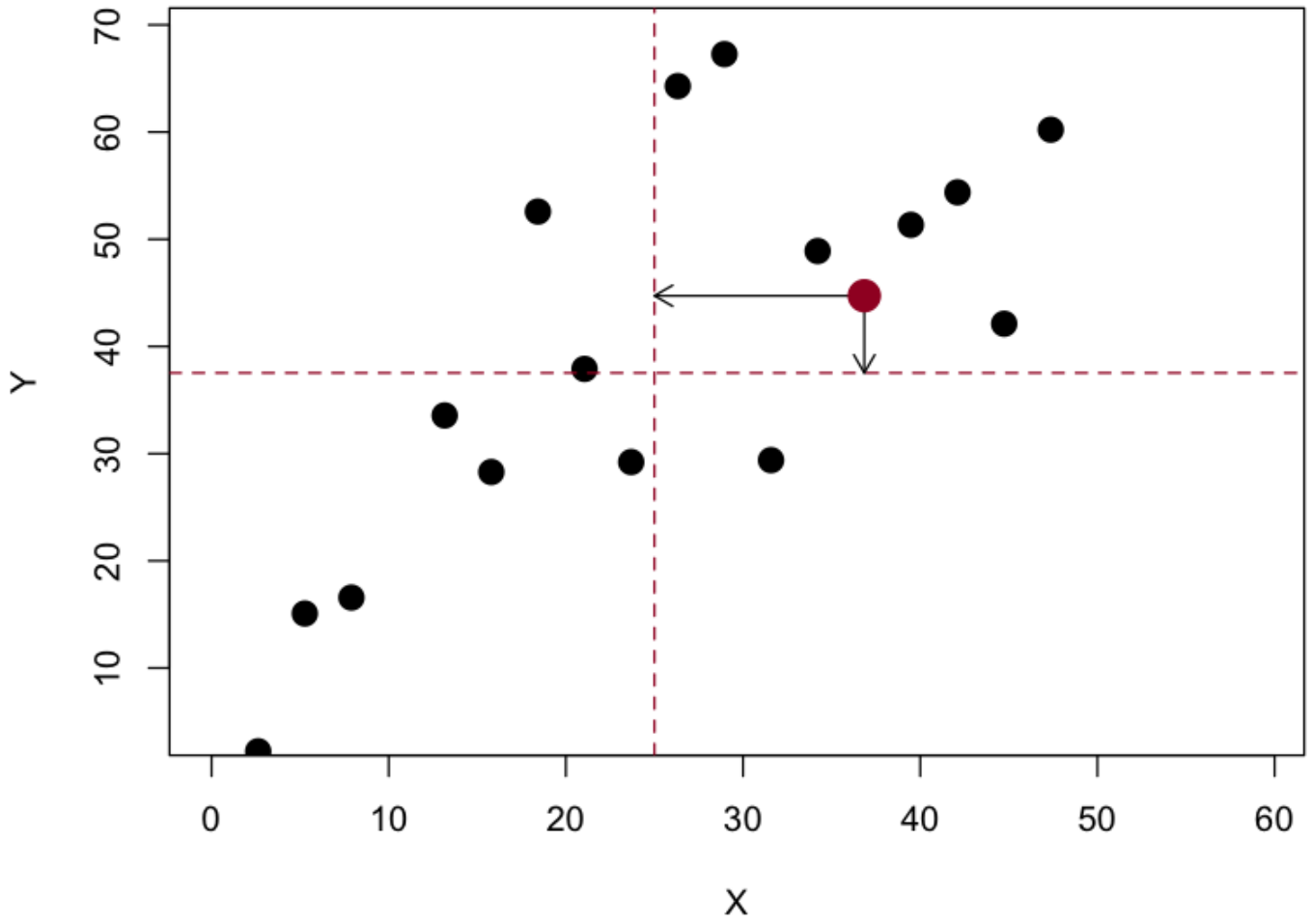
$$\sigma^2 = \frac{\Sigma(x - \bar{x})^2}{n - 1}$$

Covariance



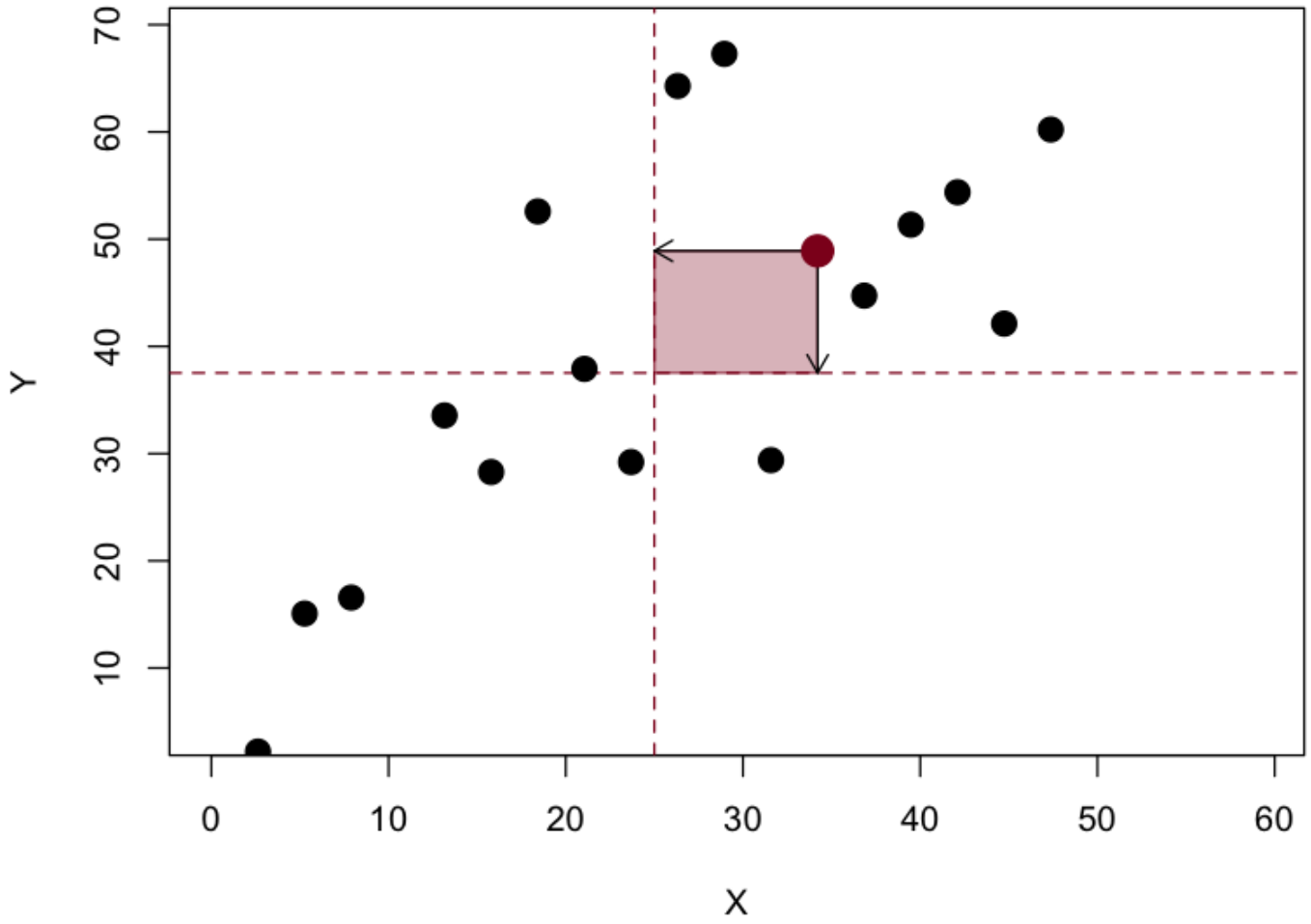
Covariance: distance from the x mean times distance from the y mean.

Covariance



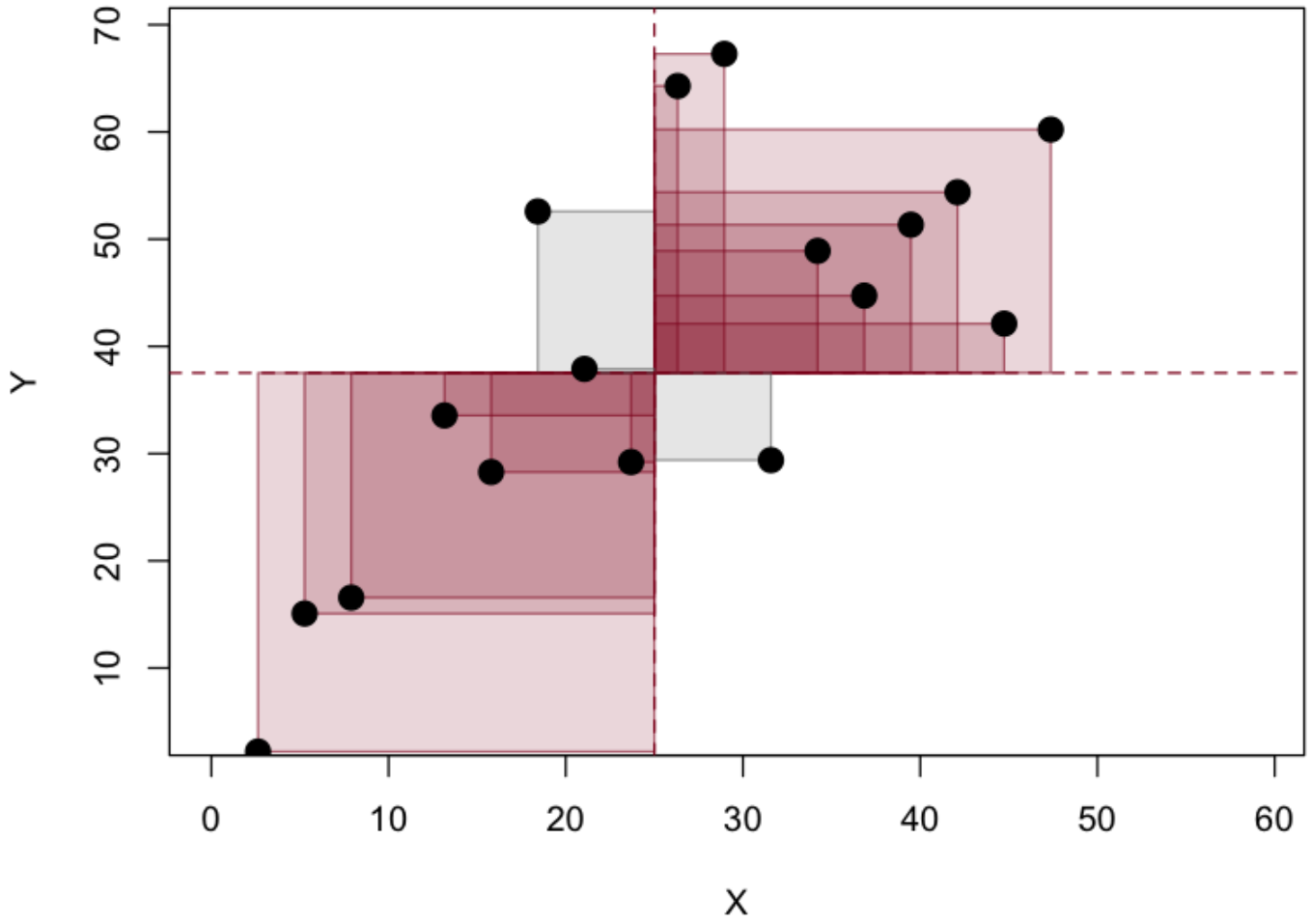
Covariance: distance from the x mean times distance from the y mean.

Covariance



The product of the two distances is the area of a rectangle.

Covariance



Dark red counts as positive, grey as negative: add up, divide by n – 1.

Covariance

- Covariance assesses the degree to which two variables change together, showing whether increases in one variable correspond to increases or decreases in another:

$$cov_{x,y} = \frac{\Sigma(x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

Correlation

- Pearson's correlation is a standardised measure that quantifies the strength and direction of the linear relationship between two variables, ranging from −1 to +1:

$$r = \frac{cov_{x,y}}{s_x s_y}$$

Dividing by the standard deviations removes the units: r is always between −1 and +1.

Correlation

Negative	Description	Positive
0.00	None	0.00
−0.19 – −0.01	Very weak	0.01 – 0.19
−0.39 – −0.20	Weak	0.20 – 0.39
−0.69 – −0.40	Modest	0.40 – 0.69
−0.89 – −0.70	Strong	0.70 – 0.89
−0.99 – −0.90	Very strong	0.90 – 0.99
−1.00	Perfect	1.00

More correlation

- Spearman's correlation is a non-parametric measure used to assess the strength and direction of the relationship between two ranked or ordinal variables.
- It is particularly useful when the data do not meet the assumptions of normality required for Pearson's correlation or when the relationship is monotonic but not linear.
- Spearman's correlation calculates the degree to which the ranks of one variable correspond to the ranks of another.

Causation

- Correlation describes the association between variables.
- Correlation, however strong, does not imply causation, but it is one important aspect of inferring causality.
- A statistically significant relationship between two variables does not mean they are causally linked.

Causation

John Stuart Mill's conditions for establishing causality are:

- Temporal precedence: The cause must come before the effect.
- Covariance: The cause and effect must be related.
- Disqualification of alternative explanations: No other variable can explain the observed relationship.

Causation

r = 0.7

Number of ice cream sales x Number of people drowning

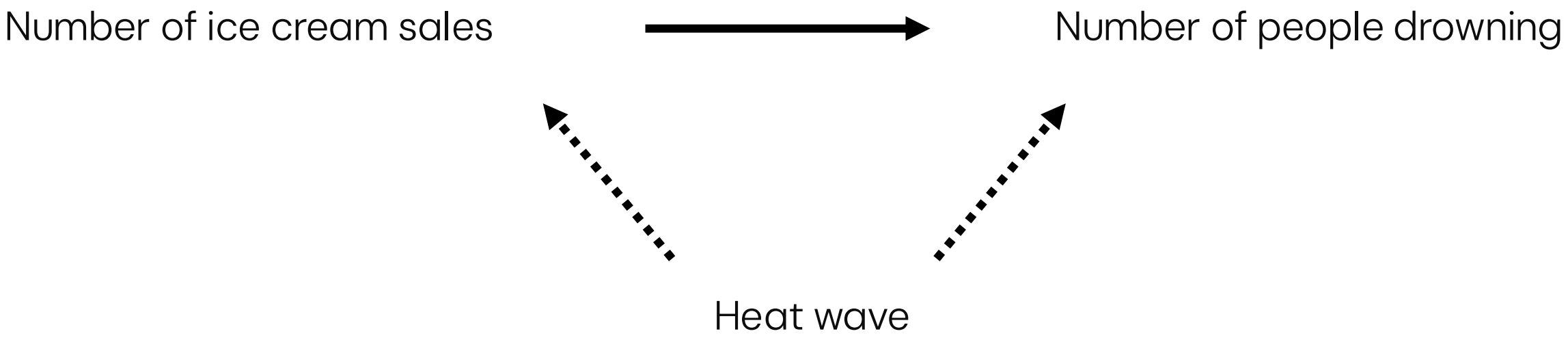
Causation

r = 0.7

Number of ice cream sales  Number of people drowning

Causation

r = 0.7



Confounder: a third variable that influences both.

Causation

r = 0.8

Number of fire fighters x Damage caused by the fire

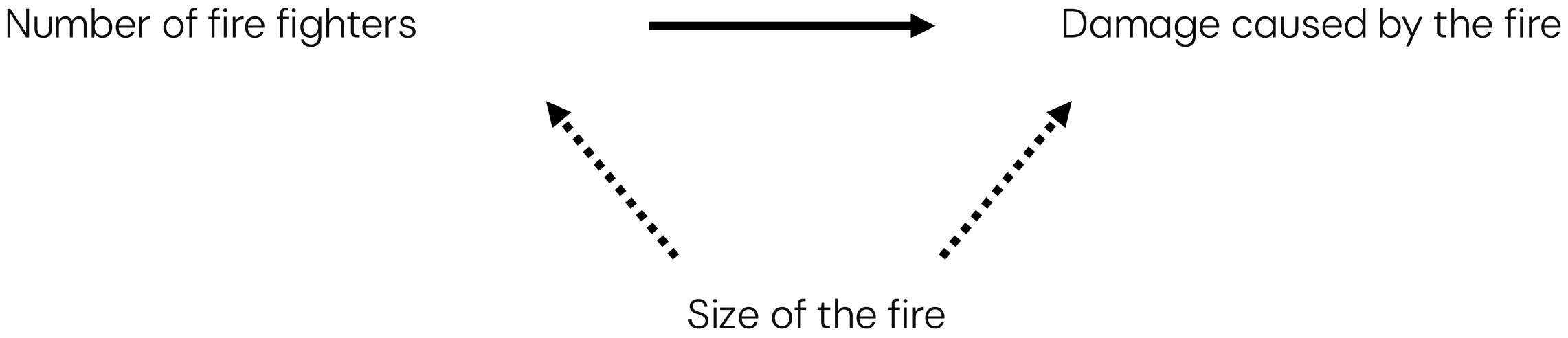
Causation

r = 0.8

Number of fire fighters  Damage caused by the fire

Causation

r = 0.8



Confounder: a third variable that influences both.

So far

- Two categorical variables: chi-square compares observed and expected counts.
- Two continuous variables: correlation shows the strength and direction of a linear relationship.
- Correlation is not causation: a third variable may drive both.

Mentimeter

- Go to www.menti.com.
- Use code: XXXX XXXX

Break

Regression

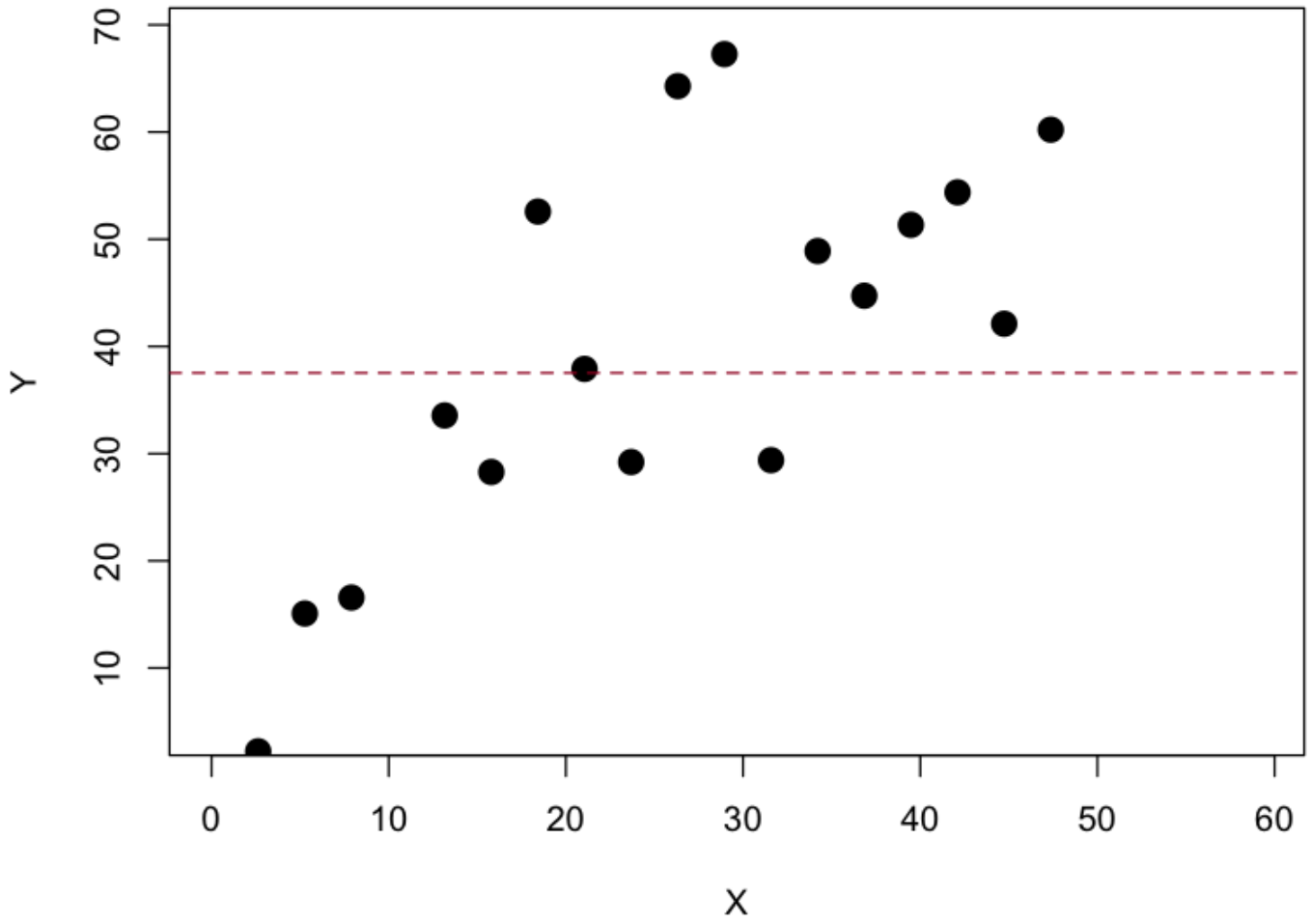
Bivariate regression

- We often want to know more than just whether two variables are related; we also want to describe and predict how one variable changes as another variable changes.
- To do this we can use a regression model to examine the relationship between a dependent variable (y) and one or more independent variables (x).

Bivariate regression

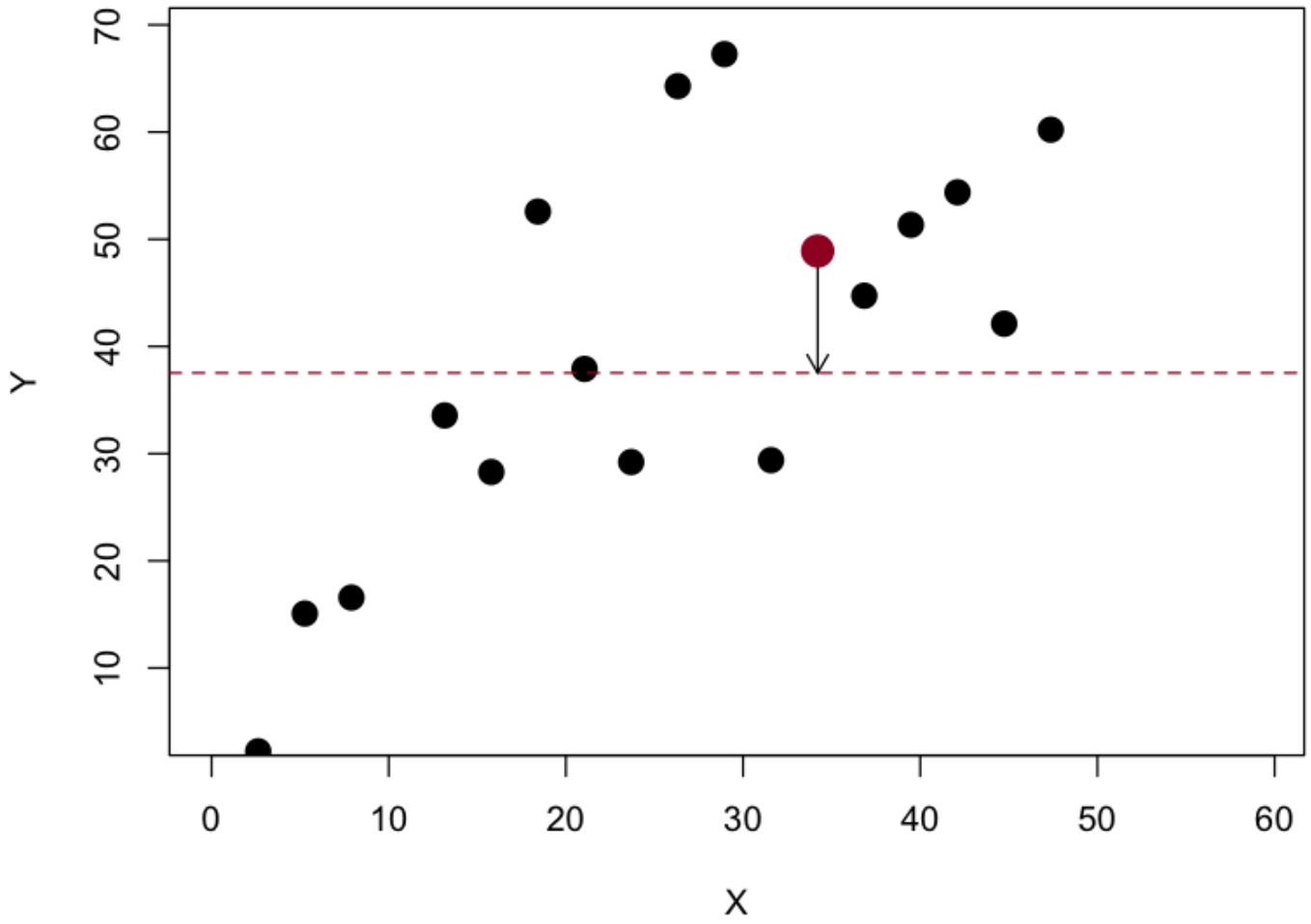
- Linear regression uses a line to summarise the relationship between x and y.
- The aim is to find the line which best represents the relationships in the data.
- Typically, this line will not pass through every data point meaning we cannot predict y exactly.

Null model



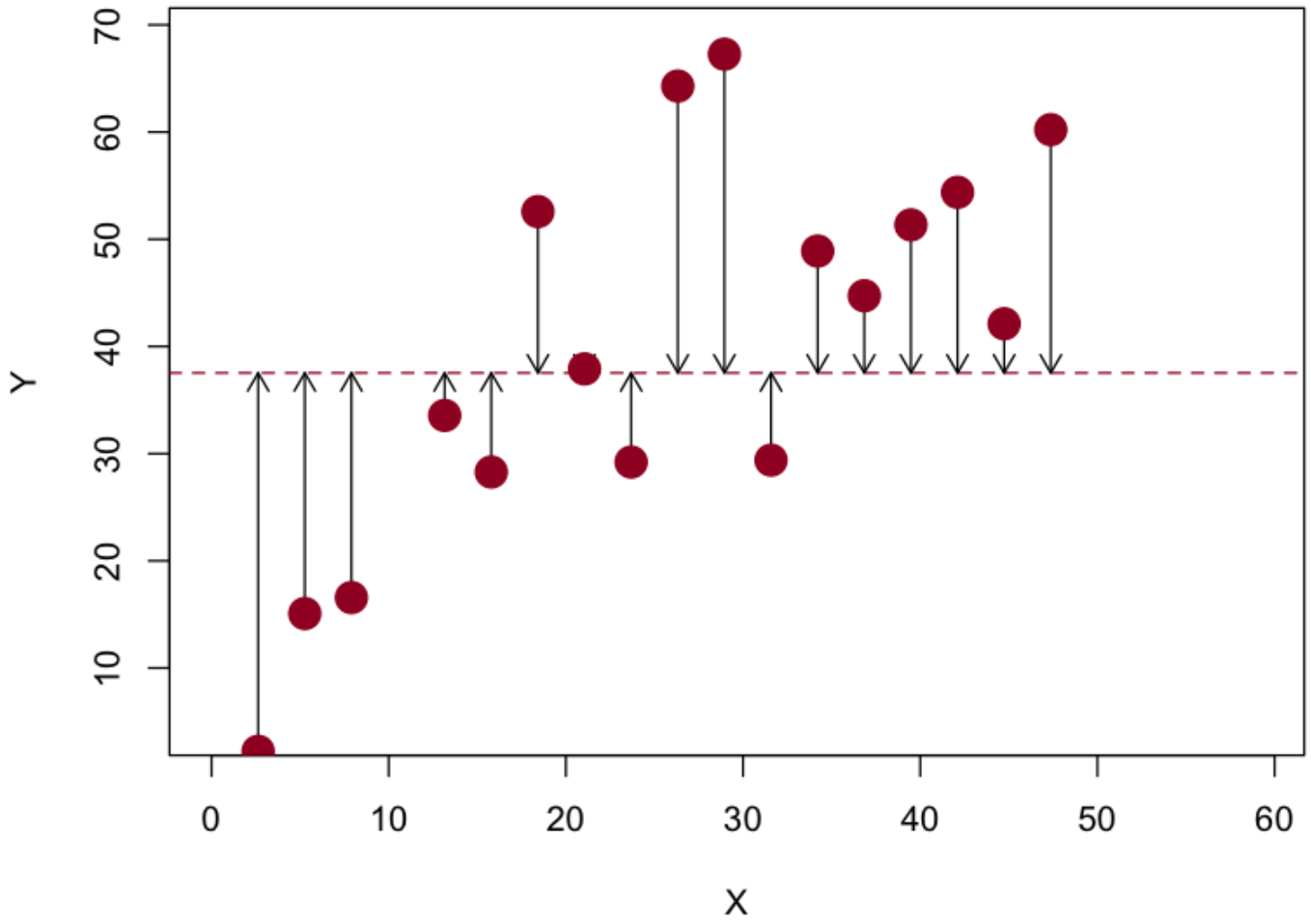
With no information about x, the best prediction of y is its mean.

Sum of squared errors



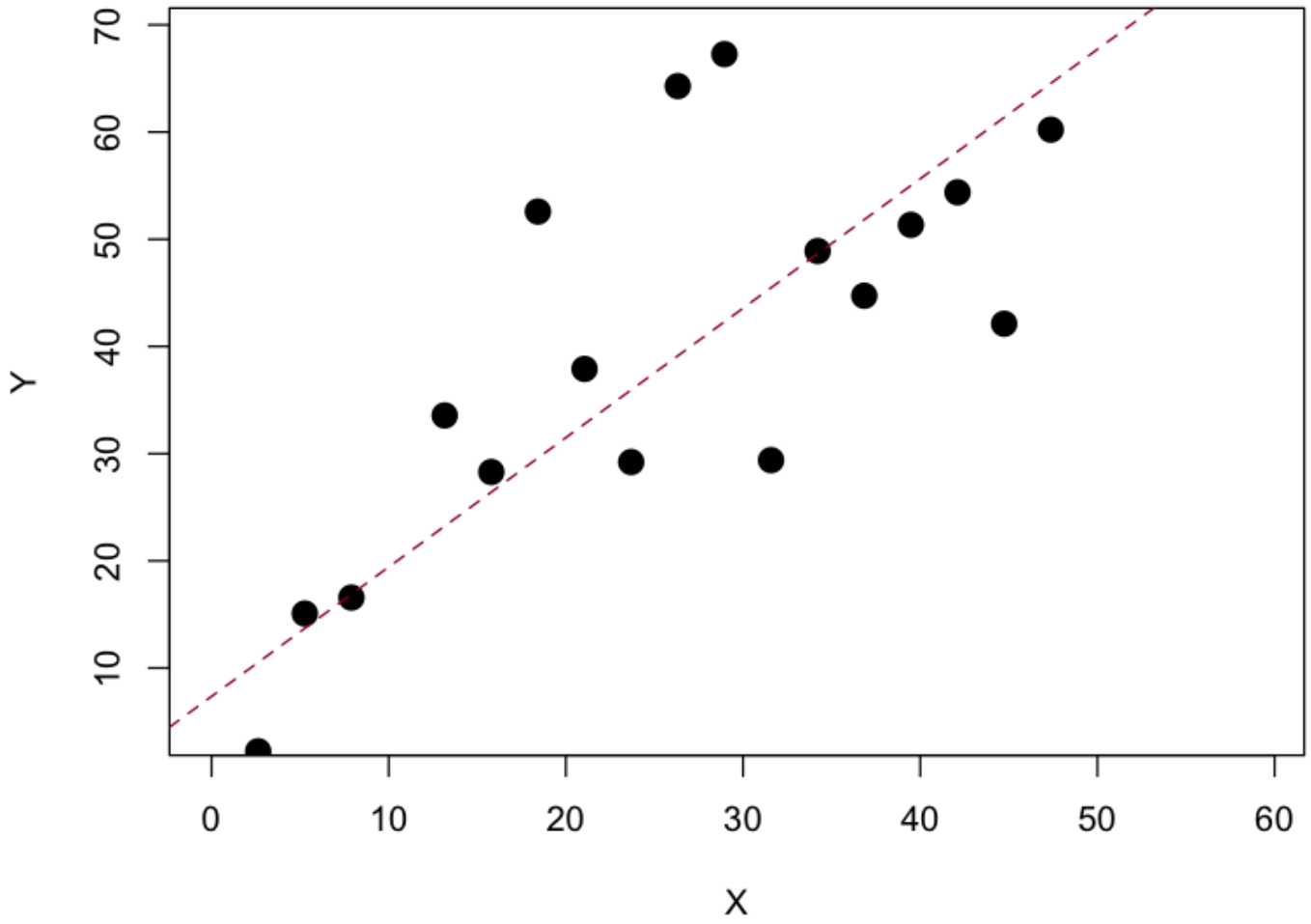
Error: the gap between an observed value and its prediction.

Sum of squared errors



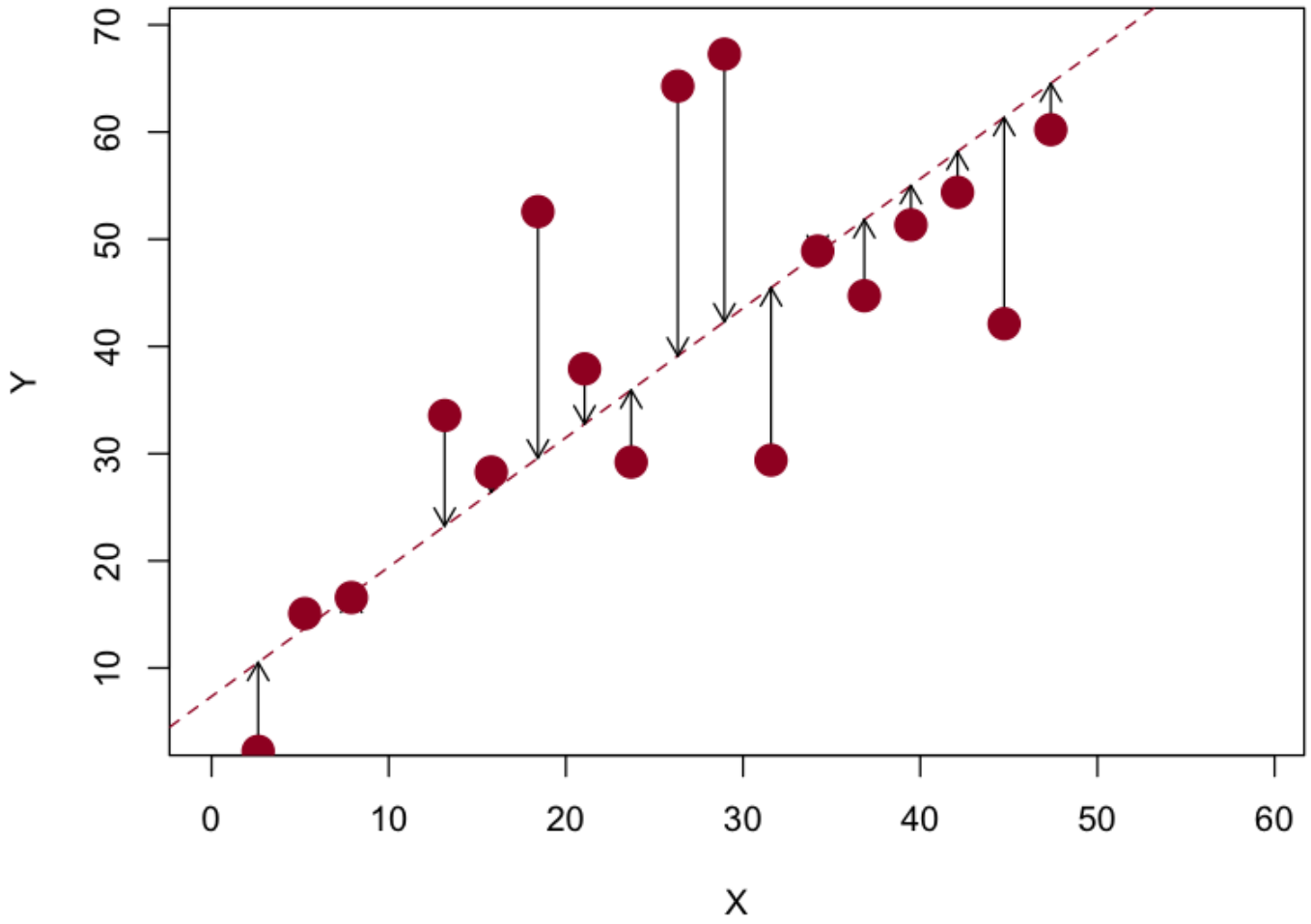
Square the errors and add them up: the sum of squared errors (SSE).

Minimise sum of squared errors



A line that follows the points leaves smaller errors.

Minimise sum of squared errors



Ordinary least squares picks the line with the smallest SSE.

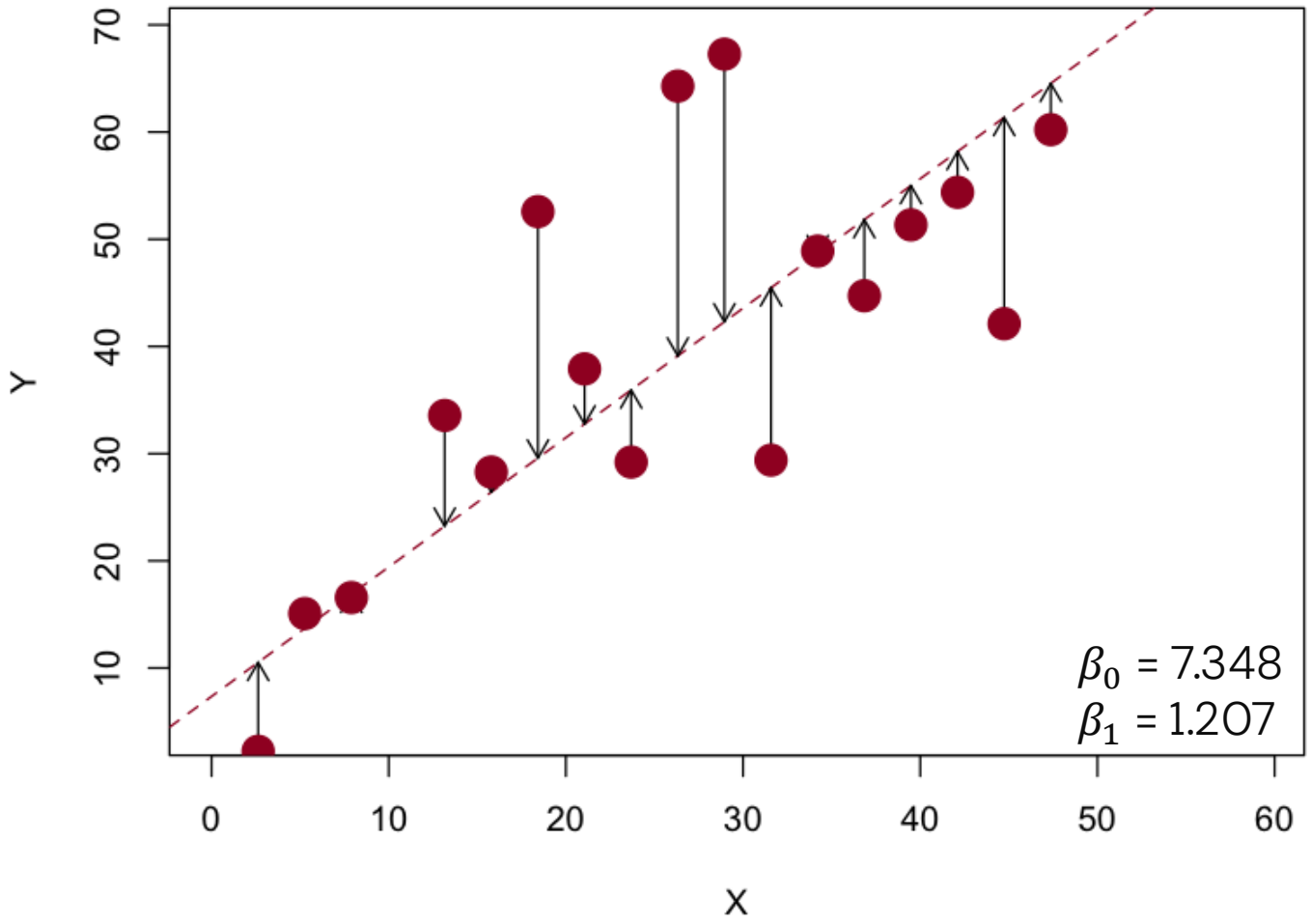
Bivariate regression

- Ordinary Least Squares (OLS) regression:

$$\hat{y} = \beta_0 + \beta_1 x$$

- The β terms are coefficients that define the regression line.
- The model estimates these parameters to find the line that gives the smallest sum of squared errors: Ordinary Least Squares (OLS) regression.

Minimise sum of squared errors



Each one-unit increase in x goes with a 1.207 increase in y, on average.

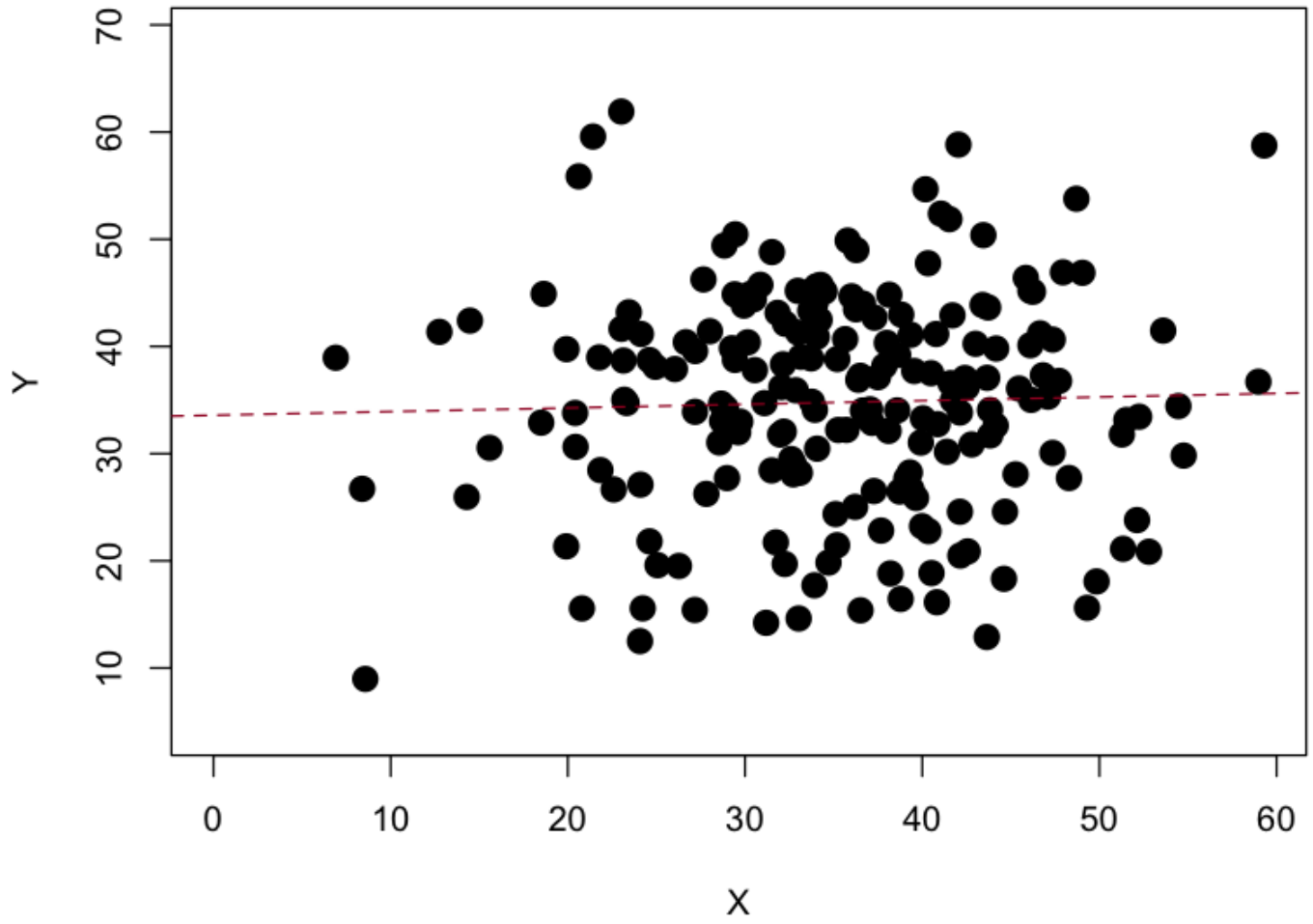
Model fit

- The R^2 measures the proportion of the variance in the dependent variable that is explained by the model.
- Values range from 0 to 1.
  - 0: The model explains none of the variance.
  - 1: The model explains all the variance.
- R^2 compares the errors of our line with the errors of the mean-only model.

Model significance

- The F statistic can be used to assess the significance of the model.
- Compares our model with a baseline model that only uses the mean: does our model fit significantly better?
- A significant F-test (p-value < 0.05) means that our model fits better than the mean-only model.
- This is the same F statistic that appeared in last week's ANOVA output.

Bivariate regression

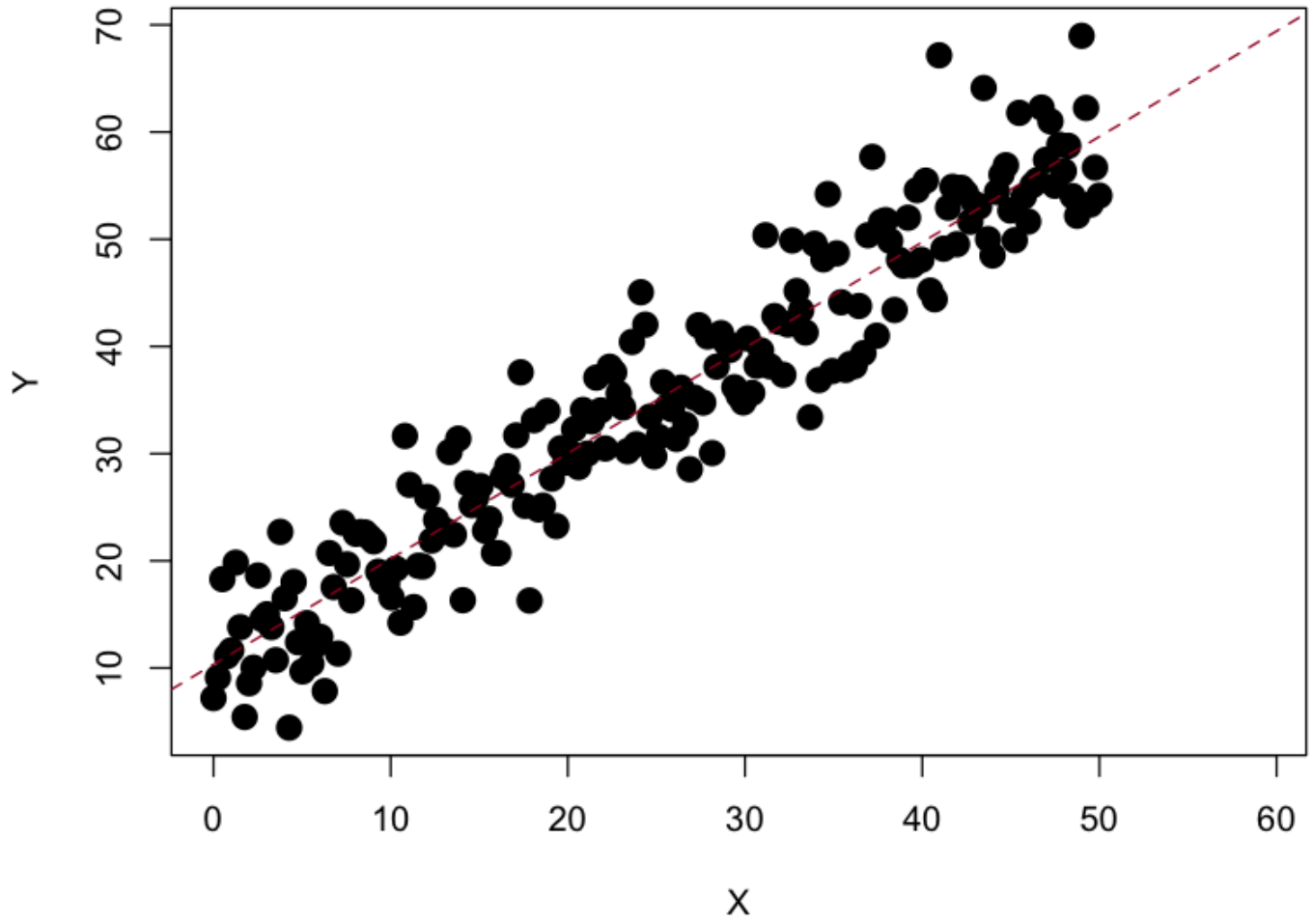


Model summary

	Dependent variable:
	y
x	0.034
	(0.075)
Constant	33.582***
	(2.761)
Observations	200
R^2	0.001
Adjusted R^2	−0.004
Residual Std. Error	10.267 (df = 198)
F Statistic	0.203 (df = 1; 198)
Note:	*p<0.1; **p<0.05; ***p<0.01

- Stars show significance. Brackets show the standard error.
- x has no stars: not significant. No clear relationship between x and y.
- R^2 = 0.001: x explains almost none of the variation in y.
- F = 0.203: no better than the mean-only model.

Bivariate regression



Model summary

	Dependent variable:
	y
x	0.985***
	(0.023)
Constant	10.340***
	(0.665)
Observations	200
R^2	0.902
Adjusted R^2	0.902
Residual Std. Error	4.722 (df = 198)
F Statistic	1,829.716*** (df = 1; 198)
Note:	*p<0.1; **p<0.05; ***p<0.01

- x: 0.985*** is significant. Each one-unit increase in x goes with a 0.985 increase in y.
- R^2 = 0.902: x explains 90% of the variation in y.
- F = 1,829.7***: far better than the mean-only model.

Multiple regression

- Multiple regression allows for a more comprehensive explanation of variance in the dependent variable compared to bivariate models.
- Multiple predictors: It is often unrealistic to attribute variations in the dependent variable y to a single factor or independent variable x.
- Separating out: Multiple regression shows how each predictor relates to the dependent variable while the other predictors are held constant.

Multiple regression

- As we add variables, the original regression equation changes:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$$

- Interpretation difference: In multiple regression, each coefficient shows the association between that predictor and y while the other predictors are held constant.

Model summary

	Dependent variable:
	y
x	0.984***
	(0.023)
z	−0.016
	(0.035)
Constant	10.907***
	(1.419)
Observations	200
R^2	0.902
Adjusted R^2	0.901
Residual Std. Error	4.732 (df = 197)
F Statistic	911.285*** (df = 2; 197)
Note:	*p<0.1; **p<0.05; ***p<0.01

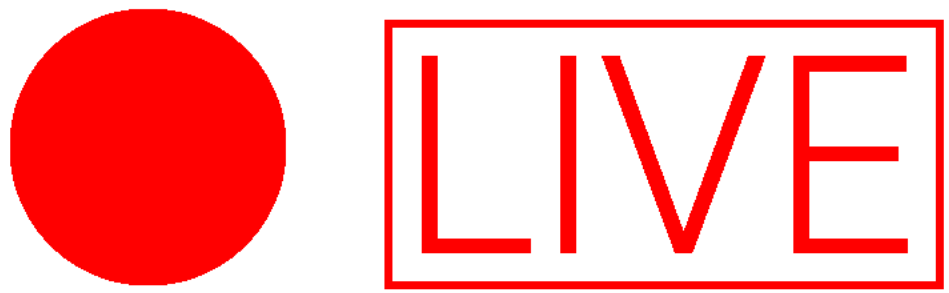
- x is significant (***). z is not: z adds nothing.
- R^2 is the same with or without z.

Adjusted R^2

- R^2 never falls when we add a variable, even a useless one.
- Adjusted R^2 takes the number of variables into account: it only rises if a new variable really helps.
- Use adjusted R^2 to compare models with different numbers of predictors.

Categorical predictors

- A categorical variable becomes dummy variables: one 0 or 1 variable for each category except the reference category.
- Example from last week: height by university, with UCL as the reference category.
- The intercept is the average height at UCL (168.3 cm). KCL is 4.9 cm taller and LSE 0.3 cm shorter than UCL.
- Last week's ANOVA is this model: the same F-test (F = 7.52).


LIVE

Assumptions

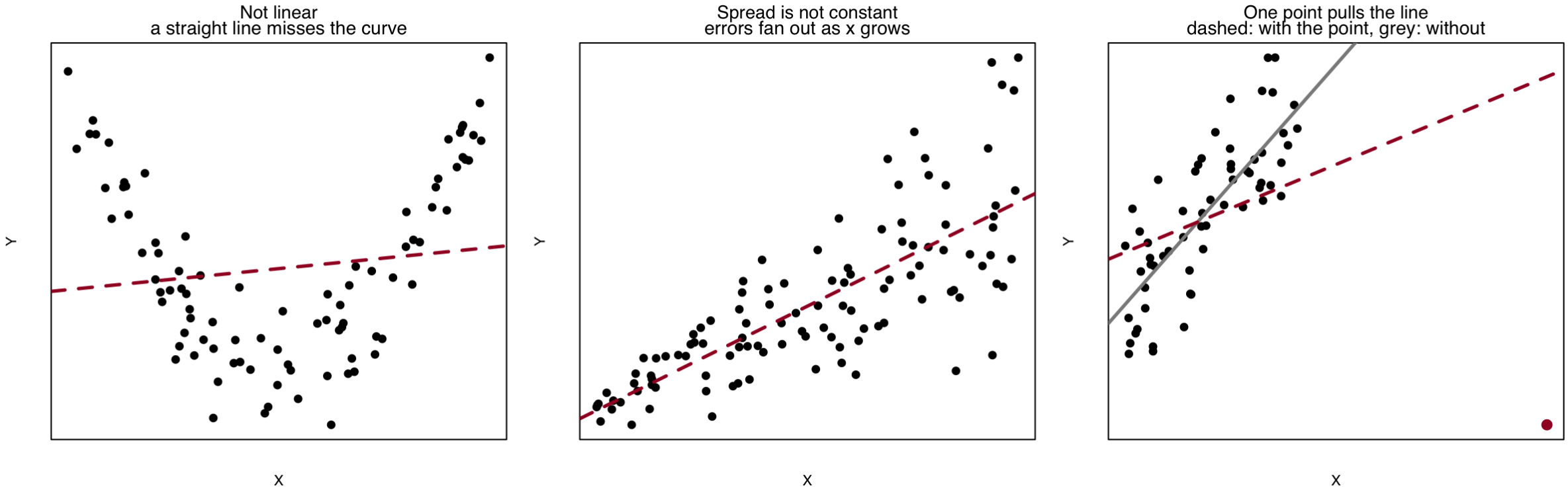
Assumptions

- There are several assumptions that must be satisfied in order to generalise our estimates beyond a sample.
- We can still use OLS when the assumptions are violated, but the estimates may be biased or inefficient.

Assumptions

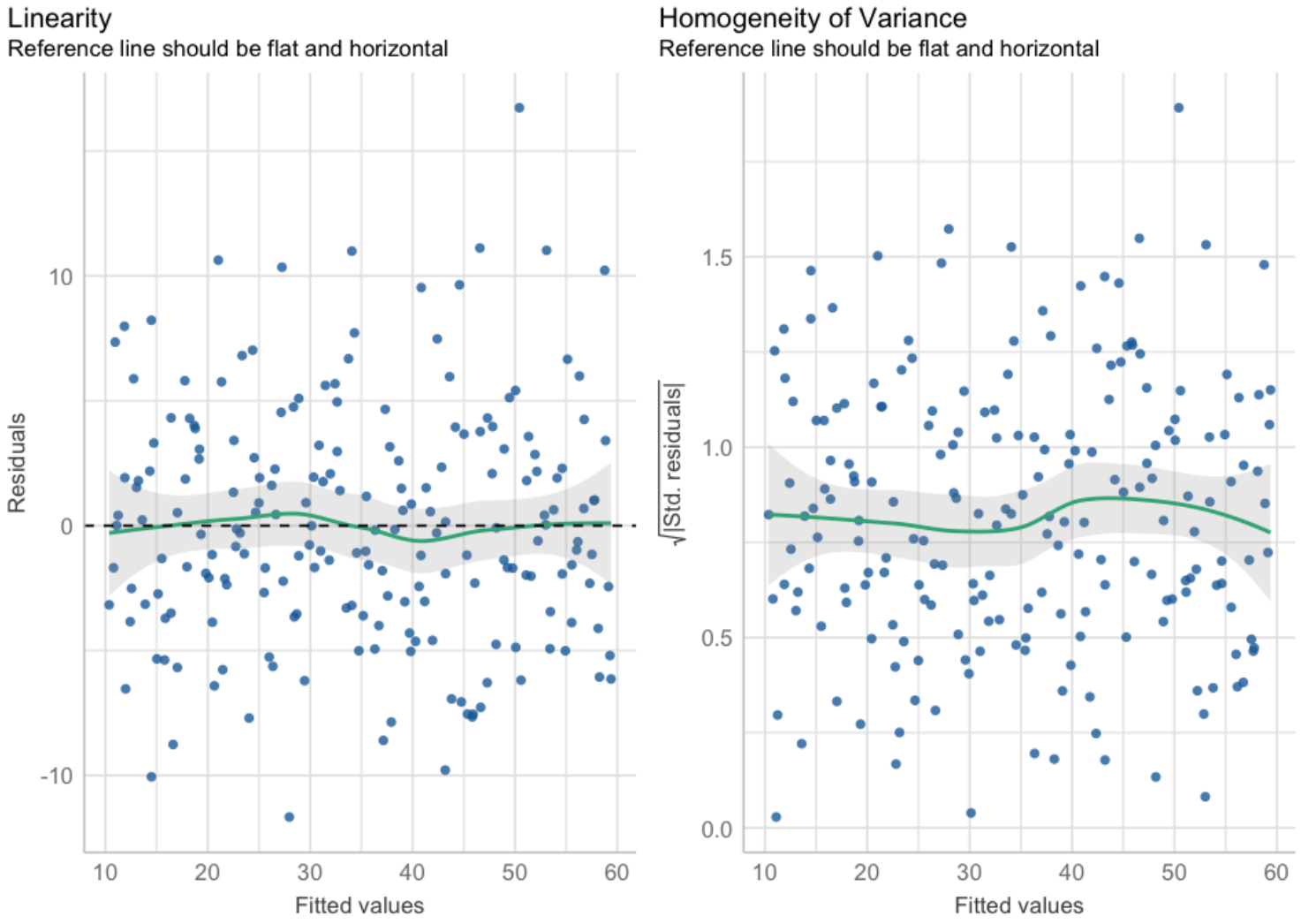
- Linearity: Each extra unit of x adds the same amount to y.
- Independence: The errors are unrelated.
- Homoscedasticity: The errors have a similar spread everywhere.
- Normality of errors: The errors are roughly bell-shaped.
- No multicollinearity: The predictors are not strongly correlated with each other.
- No influential observations: No single place drives the result.

Assumptions: what a problem looks like



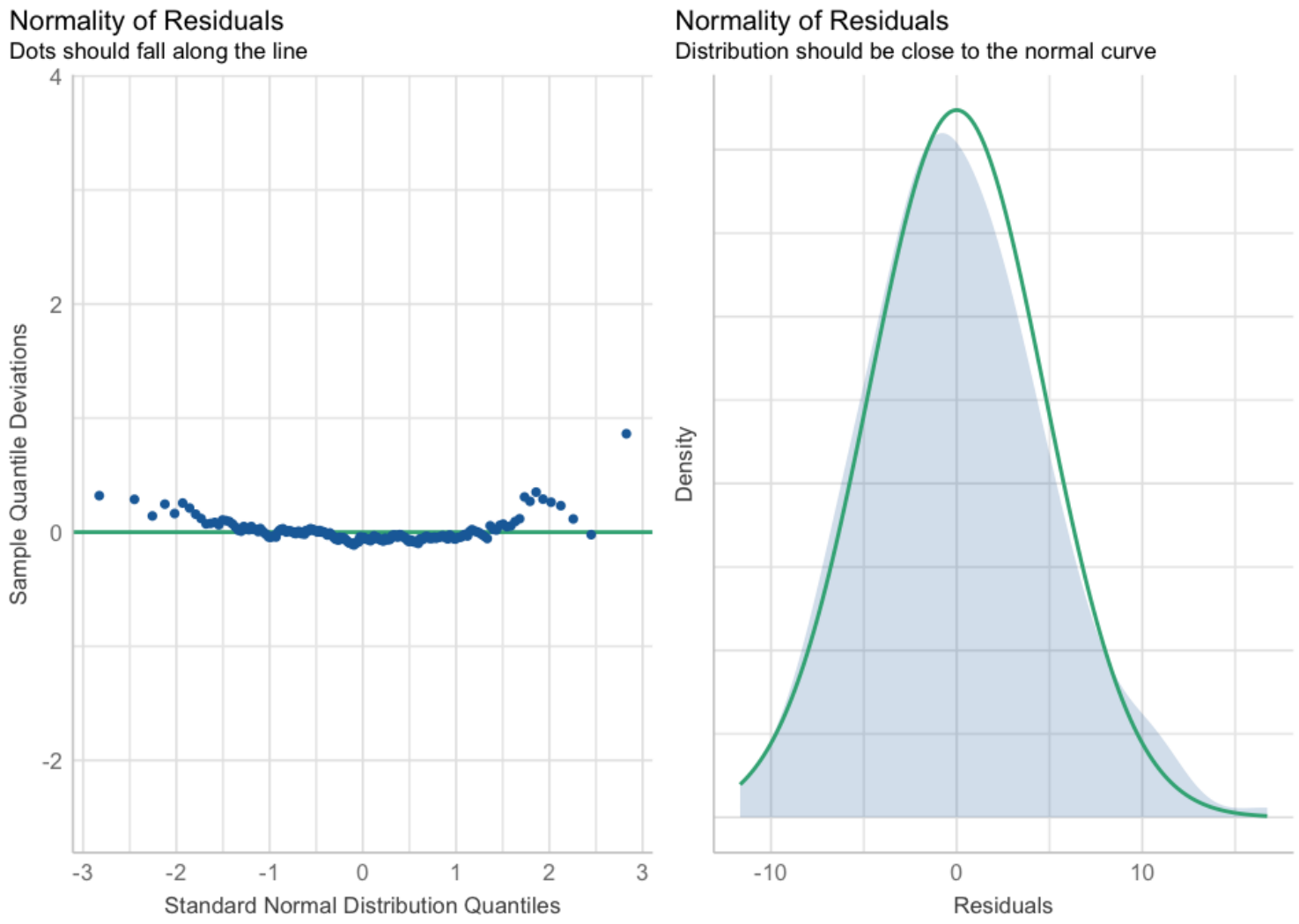
Plot the data and the errors: patterns like these tell us when to worry.

Assumptions



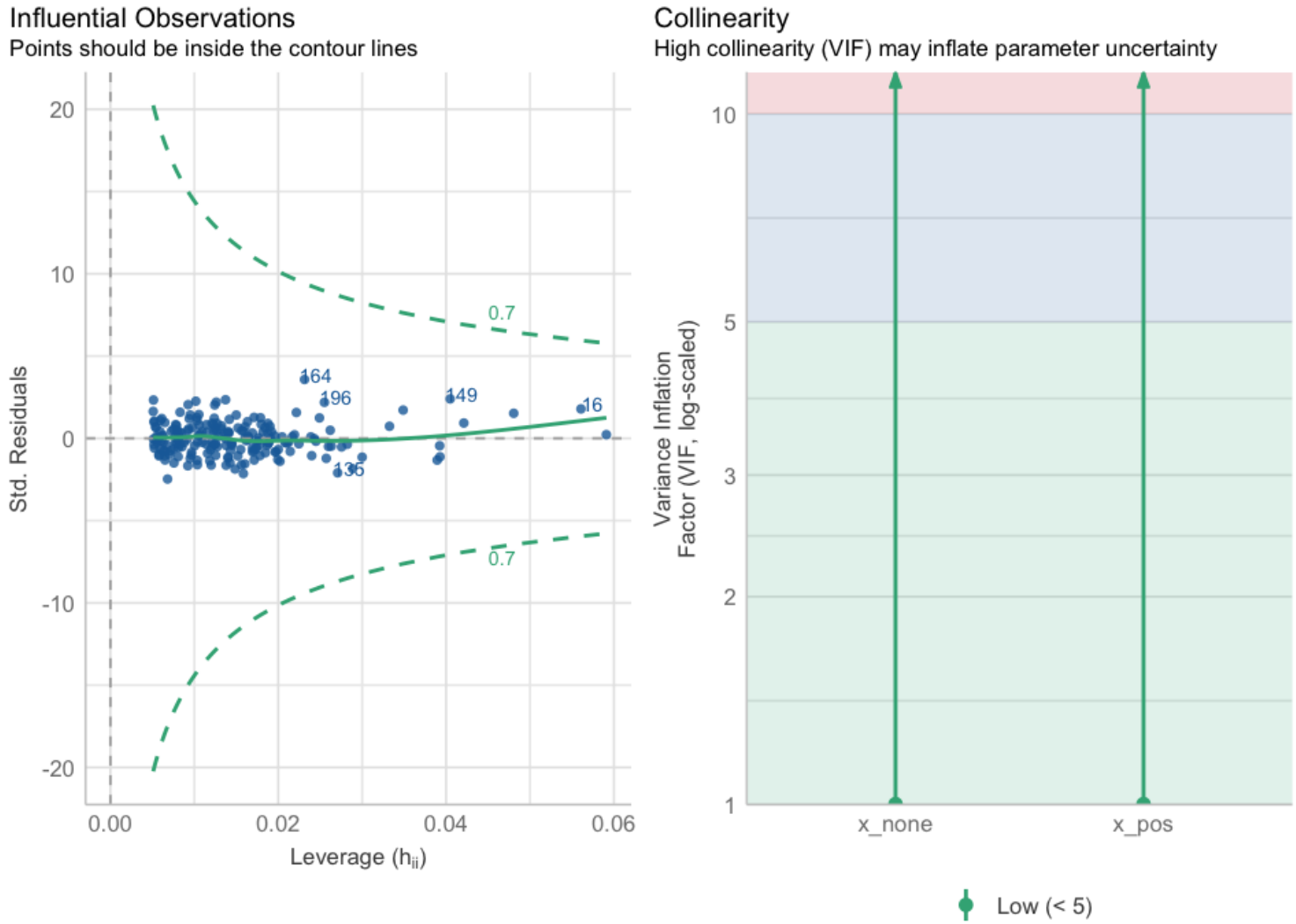
Output from the *easystats* R package.

Assumptions



Output from the *easystats* R package.

Assumptions



Output from the *easystats* R package.

Multicollinearity

- Multicollinearity means that predictors are strongly correlated with each other.
- This makes the coefficients unstable and hard to interpret.
- The variance inflation factor (VIF) measures it: below 5 is fine, 5 to 10 is moderate, above 10 is a problem.

More regression?

- OLS regression on non-linear patterns (like curves): interaction terms, polynomials.
- Logistic regression to model the relationship between a binary dependent variable and one or more independent variables, estimating the probability of the outcome occurring.
- Poisson regression to model count data, where the dependent variable represents the number of events occurring within a fixed interval, assuming the events occur independently.

More regression?

- Time series regression for using historical data points collected over time to predict future values of a dependent variable, taking into account trends, seasonality, and autocorrelation.
- Multilevel regression to account for data with hierarchical or nested structures, allowing for the analysis of relationships at multiple levels.
- Geographically Weighted Regression to analyse spatial variations in relationships between variables by estimating coefficients for each observation based on its geographic location.

Summary

Summary

- Crosstabulation and chi-square: the association between two categorical variables.
- Correlation: the strength and direction of a linear relationship. Regression: how y changes with x.
- Read a regression through its coefficients, R^2 and F-test, and check the assumptions.
- These methods describe associations, not causes.

Questions

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